

PIPING ARRANGEMENT EDIT

Arrangement Edit (SP3D)

SmartPlant 3D arrangement 100-series scripts: centerline, symbol, line-no, dimension, tag arrange.

Key Purpose

Why this program

1. Turn raw SP3D-exported DXF into a finished GA drawing

- One click chains 10 steps: centerline, symbols, line-no, dimensions, tags, masking

→ **Effect:** Hours of manual CAD clean-up per arrangement drawing collapse into a single automatic run

2. Auto-detect drawing options from the layers/text

- Reads SP3D layer names and text to preset elevation mode, scale, and which tags exist

→ **Effect:** The dialog opens already configured, so operators rarely need to guess or set anything by hand

3. Standardize elevation, tags and annotation styling

- Rewrites BOP/COP/TOP + EL/FL/GL elevations and applies consistent tag colors, sizes, fonts

→ **Effect:** Every drawing follows the same house style with correct pipe-elevation math, cutting review markups

4. Batch process many drawings at once

- Select a whole list of DXF files; each is processed and saved as _ALL.DXF

→ **Effect:** A full package of arrangement sheets is regenerated unattended instead of one file at a time

Required inputs : DXF arrangement drawings exported from SmartPlant 3D (SP3D)

Main Window – Item Guide

General Tab

The screenshot shows the 'Piping GA Drawing Option for SP3D' dialog box. It has three tabs: 'General' (selected), 'Annotation', and 'Dimension'. The 'General' tab contains the following elements:

- 1**: Tab bar area.
- 2**: 'Select Dxf Arrangement Dwg From SP3D' section with a file list box.
- 3**: 'Select', 'Remove Selected', 'Clear All', and 'Open' buttons.
- 4**: 'Elevation Expression' section with 'BOP' and 'TOP' buttons.
- 5**: 'EL or FL or GL' section with 'EL.' and 'GL.' buttons.
- 6**: 'Nozzle Table Create' section with 'Nozzle Table' and 'Nozz No Annotation' checkboxes.
- 7**: 'Tagging Option From SP3D Draft Data' section with checkboxes for 'Line No Annotation', 'Valv No Annotation', 'Inst No Annotation', 'Support Annotation', 'Flow Mark', 'Equip Annotation', 'Stru Annotation', 'WP EL', 'Redu Annotation', and 'Tie Point Annotation'.
- 8**: 'Redu Annotation' checkbox.
- 9**: 'All Dwg Text Font' dropdown menu set to 'Arial'.
- 10**: 'Apply' button.
- 11**: 'Save' button.
- 12**: 'Load' button.

1. General / Annotation / Dimension option tabs
2. Selected SP3D DXF arrangement file list
3. Select / Remove Selected / Clear All / Open
4. Elevation Expression: BOP / COP / TOP
5. Datum: EL. / FL. / GL.
6. Nozzle Table Create checkboxes
7. Tagging left: Line / Valv / Inst / Support / Flow
8. Tagging right: Equip / Stru / WP EL / Redu / Tie
9. All Dwg Text Font selector
10. Apply — run the 000-108 processing chain
11. Save / Load option preset
12. Download selected drawings

Main Window – Item Guide

Annotation Tab

Piping GA Drawing Option for SP3D

General **Annotation** Dimension

Line No Annotation

☒ Line Cyan **2** Text Yellow **3**

Text Size 2.5 Line No Overflow Off **4**

☐ Min Line no : Left, Up/Down, Low /Right, Up/Right, Low **5**

Equip Annotation

☒ Line Cyan **6** Text Cyan

Text Size 3 Symbol Type Default

Stru. Column No Annotation

☒ Line Cyan **7** Text Yellow

Text Size 4.5 Circle Size 13

Valv No Annotation

☒ Line Cyan **8** Text Yellow

Text Size 2 Symbol Type Rhombus

Inst No Annotation

☒ Line Cyan **9** Text Yellow

Text Size 2 Symbol Type Circle

WP EL.

Apply **10** Save **12** Load Download

1. General / Annotation / Dimension option tabs
2. Line No: Line color combo
3. Line No: Text color combo
4. Line No: Text Size + Line No OverFlow
5. Min Line no corner-keep checkbox
6. Equip Annotation: color / size / Symbol Type
7. Stru Column No: Text Size + Circle Size
8. Valv No Annotation (Symbol Type Rhombus)
9. Inst No Annotation (Symbol Type Circle)
10. WP EL. block (scroll for Support/Redu/Tie)
11. Vertical scrollbar to more tag blocks
12. Apply / Save / Load / Download footer

Main Window – Item Guide

Dimension Tab

The screenshot shows the 'Piping GA Drawing Option for SP3D' window with the 'Dimension' tab selected. The interface includes several sections for configuring drawing options, with numbered callouts (1-12) highlighting specific elements:

- 1**: General / Annotation / Dimension option tabs
- 2**: Unit selector (Millimetres)
- 3**: Arrow end Type selector
- 4**: Dim. Leader Line style (Dotx2)
- 5**: Dup. Line Dim threshold (DN50)
- 6**: Piping Dimension: Line/Text + Gap&Size
- 7**: Equipment Dimension: Line/Text + Gap&Size
- 8**: Structure Dimension: Line/Text + Gap&Size
- 9**: Expression Dimension: Left/Right/Up/Down
- 10**: Expression Stru. Col. No sides
- 11**: Dwg Limit: L+U / L+D / R+U / R+D
- 12**: Apply / Save / Load / Download footer

1. General / Annotation / Dimension option tabs
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Processing Steps

What 'Apply' runs, in order

1. 000 Create pipe centerlines from the SP3D geometry
2. 100 Apply file settings and standard layers
3. 101 Edit and standardize piping symbols
4. 102 Delete duplicate and overflow line numbers
5. 103 Stretch and reposition line-number leaders
6. 104 Create dimensions
7. 105 Edit and clean up dimensions
8. 106 Edit tag numbers (line / valve / inst / equip / stru)
9. 107 Arrange and de-clutter the tag numbers
10. 108 Mask tags where they overlap the drawing
11. Rewrite pipe elevations (BOP/COP/TOP + EL/FL/GL) with OD math
12. Apply annotation colors and text sizes
13. Save each drawing as <name>_ALL.DXF

Overview

What it does

1. SmartPlant 3D arrangement 100-series scripts: centerline
2. Symbol
3. Line-no
4. Dimension
5. Tag arrange

Category: Piping Arrangement Edit

Folder: PIPING_ARRANGEMENT_EDIT/SP3D

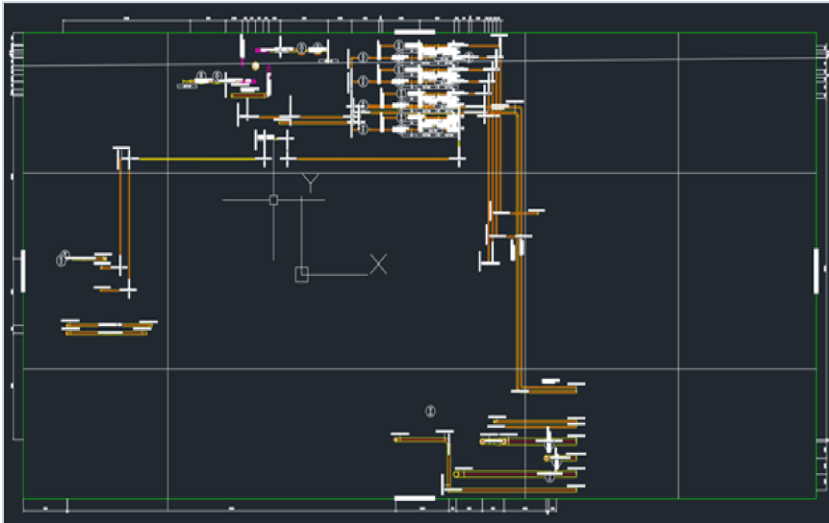
Main scripts: (component scripts)

Scripts / Status: 14 scripts · Operational

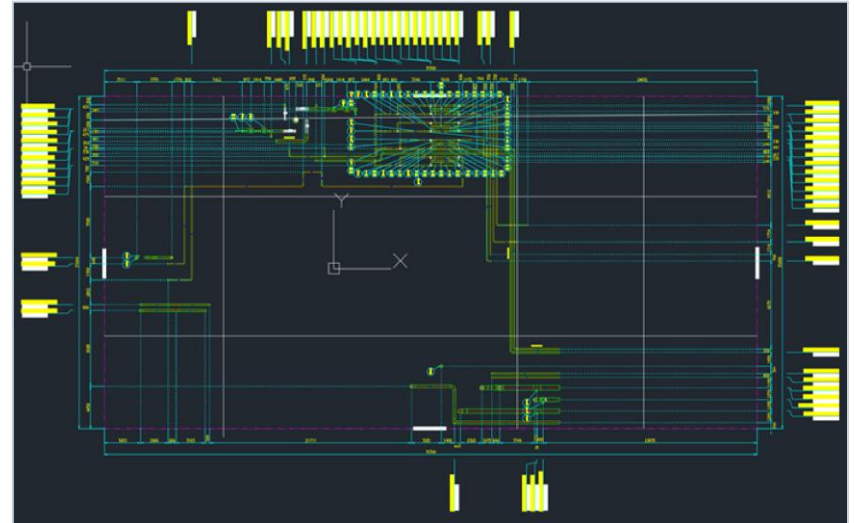
Tags: SP3D · 평면도 · 100번대

Output Samples

Output Samples



SP3D Input File for Piping GA



SP3D Auto Output file for Piping GA

Notes & Information

- Part of the in-house Plant Engineering automation suite — category: Piping Arrangement Edit.
- Runs offline as a pure-Python / Tkinter tool; portable across PCs via OneDrive sync (needs Python + packages).
- A set of component scripts — run the relevant step from the launcher / script list.
- Output samples and the program window above reflect the current build.

Thank You

Arrangement Edit (SP3D) · Plant Engineering Total Service System